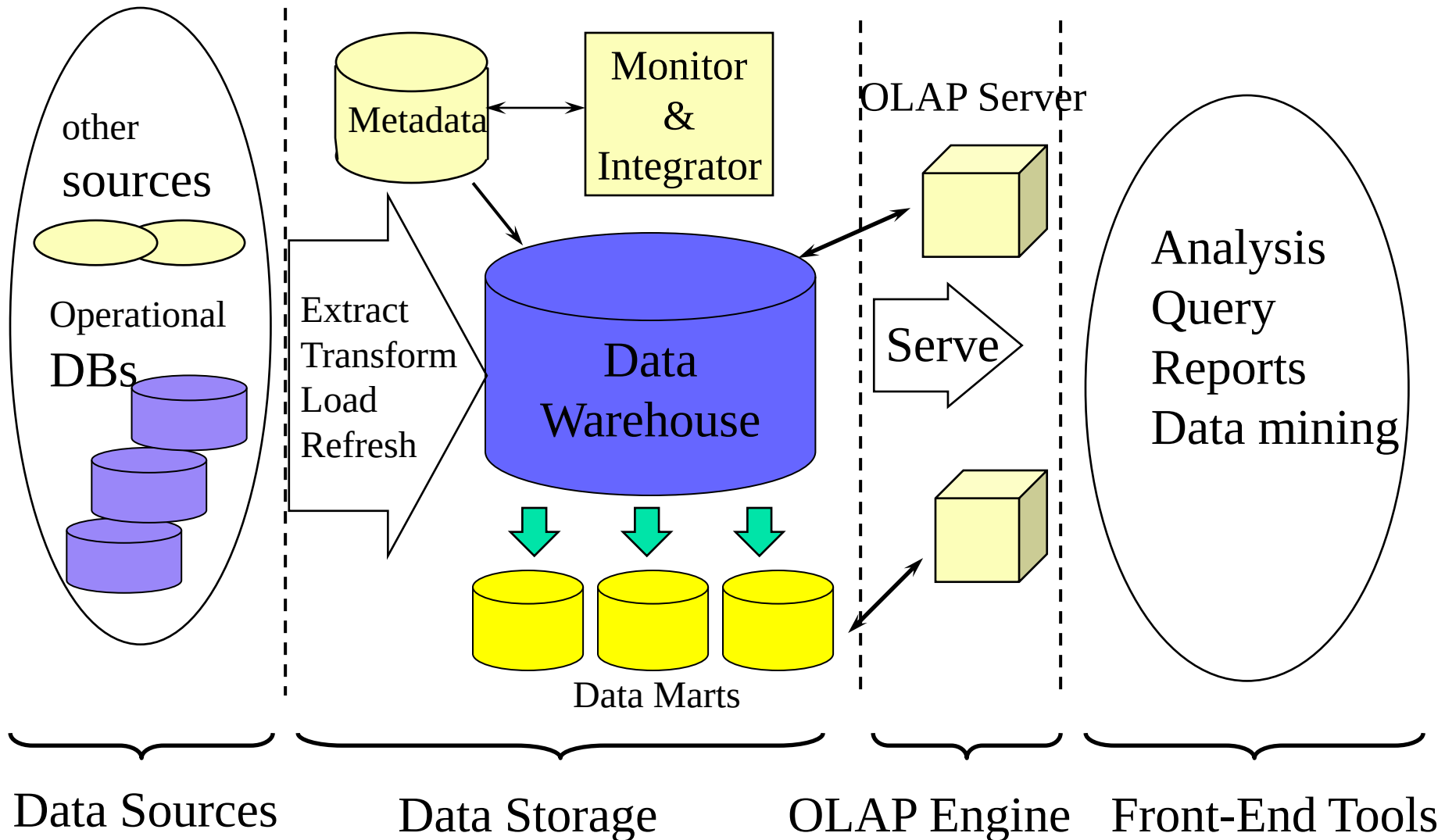


What is Data Warehouse?

- Defined in many different ways, but not rigorously.
 - A decision support database that is maintained **separately** from the organization's operational database
 - Support **information processing** by providing a solid platform of consolidated, historical data for analysis.
- "A data warehouse is a subject-oriented, integrated, time-variant, and nonvolatile collection of data in support of management's decision-making process."—W. H. Inmon
- Data warehousing:
 - The process of constructing and using data warehouses

Data Warehouse Architecture



Data Warehouse—Subject-Oriented

- Organized around major subjects, such as **customer, product, sales**.
- Focusing on the modeling and analysis of data for decision makers, not on daily operations or transaction processing.
- Provide **a simple and concise** view around particular subject issues by **excluding data that are not useful in the decision support process**.

Data Warehouse—Integrated

- Constructed by integrating multiple, heterogeneous data sources
 - relational databases, flat files, on-line transaction records
- Data cleaning and data integration techniques are applied.
 - Ensure consistency in naming conventions, encoding structures, attribute measures, etc. among different data sources
 - E.g., Hotel price: currency, tax, breakfast covered, etc.
 - When data is moved to the warehouse, it is converted.

Data Warehouse—Time Variant

- The time horizon for the data warehouse is significantly longer than that of operational systems.
 - Operational database: current value data.
 - Data warehouse data: provide information from a historical perspective (e.g., past 5-10 years)
- Every key structure in the data warehouse
 - Contains an element of time, explicitly or implicitly
 - But the key of operational data may or may not contain “time element”.

Data Warehouse—Non-Volatile

- A **physically separate store** of data transformed from the operational environment.
- Operational **update of data does not occur** in the data warehouse environment.
 - Does not require transaction processing, recovery, and concurrency control mechanisms
 - Requires only two operations in data accessing:
 - *initial loading of data* and *access of data*.

Data Warehouse vs. Operational DBMS

- OLTP (on-line transaction processing)
 - Major task of traditional relational DBMS
 - Day-to-day operations: purchasing, inventory, banking, manufacturing, payroll, registration, accounting, etc.
- OLAP (on-line analytical processing)
 - Major task of data warehouse system
 - Data analysis and decision making
- Distinct features (OLTP vs. OLAP):
 - User and system orientation: customer vs. market
 - Data contents: current, detailed vs. historical, consolidated
 - Database design: ER + application vs. star + subject
 - View: current, local vs. evolutionary, integrated
 - Access patterns: update vs. read-only but complex queries

OLTP vs. OLAP

	OLTP	OLAP
users	clerk, IT professional	knowledge worker
function	day to day operations	decision support
DB design	application-oriented	subject-oriented
data	current, up-to-date detailed, flat relational isolated	historical, summarized, multidimensional integrated, consolidated
usage	repetitive	ad-hoc
access	read/write index/hash on prim. key	lots of scans
unit of work	short, simple transaction	complex query
# records accessed	tens	millions
#users	thousands	hundreds
DB size	100MB-GB	100GB-TB
metric	transaction throughput	query throughput, response

Why Separate Data Warehouse?

- High performance for both systems
 - DBMS—tuned for OLTP: access methods, indexing, concurrency control, recovery
 - Warehouse—tuned for OLAP: complex OLAP queries, multidimensional view, consolidation.
- Different functions and different data:
 - missing data: Decision support requires historical data which operational DBs do not typically maintain
 - data consolidation: DS requires consolidation (aggregation, summarization) of data from heterogeneous sources
 - data quality: different sources typically use inconsistent data representations, codes and formats which have to be reconciled

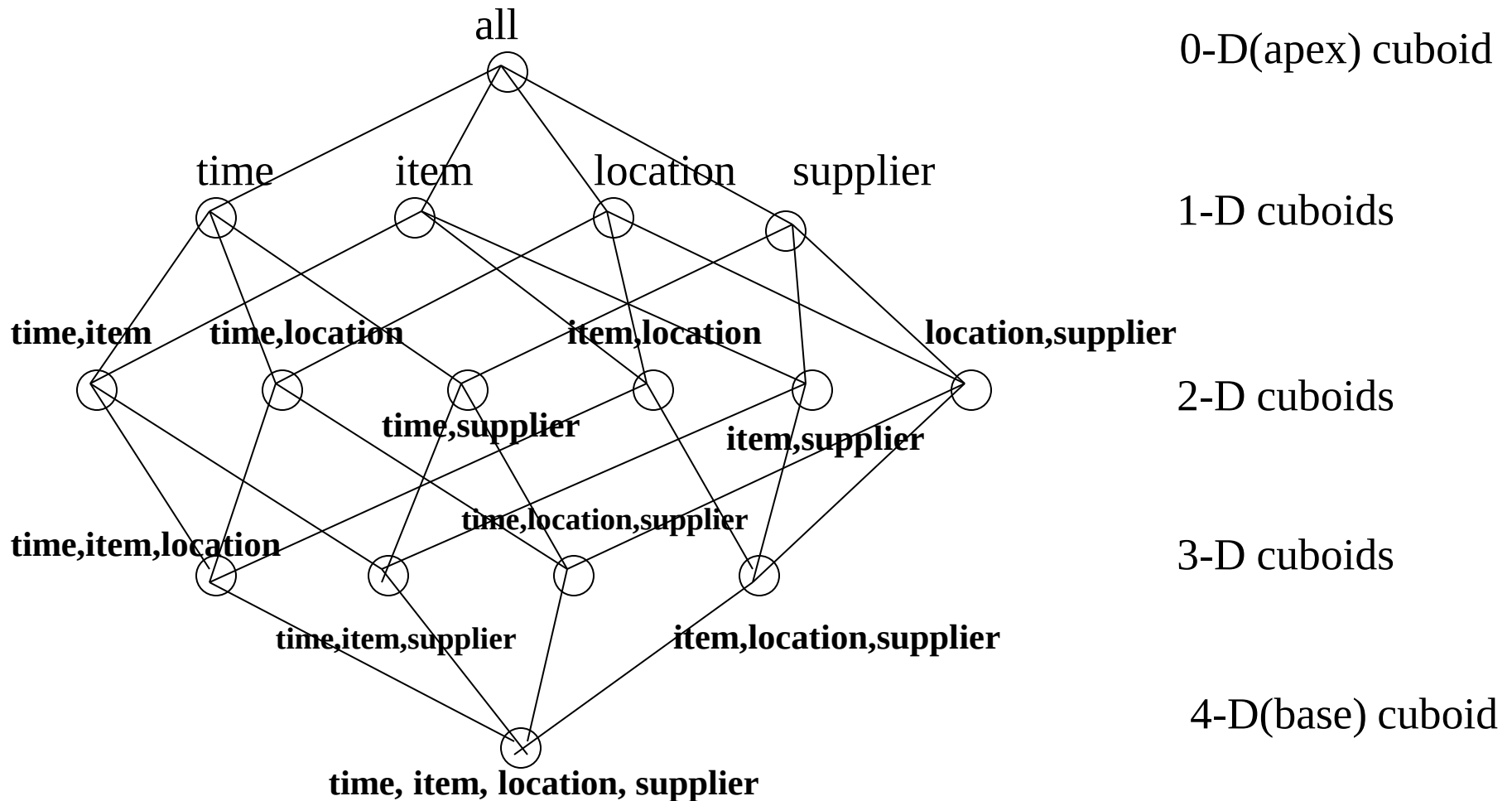
Data Warehousing and OLAP Technology for Data Mining

- What is a data warehouse?
- A multi-dimensional data model
- Data warehouse architecture
- Data warehouse implementation
- Further development of data cube technology
- From data warehousing to data mining

From Tables and Spreadsheets to Data Cubes

- A data warehouse is based on a **multidimensional data model** which views data in the form of a data cube
- A data cube, such as **sales**, allows data to be modeled and viewed in multiple dimensions
 - Dimension tables, such as **item (item_name, brand, type)**, or **time(day, week, month, quarter, year)**
 - Fact table contains measures (such as **dollars_sold**) and keys to each of the related dimension tables
- In data warehousing literature, an n-D base cube is called a **base cuboid**. The top most 0-D cuboid, which holds the highest-level of summarization, is called the **apex cuboid**. The lattice of cuboids forms a **data cube**. (stop here)

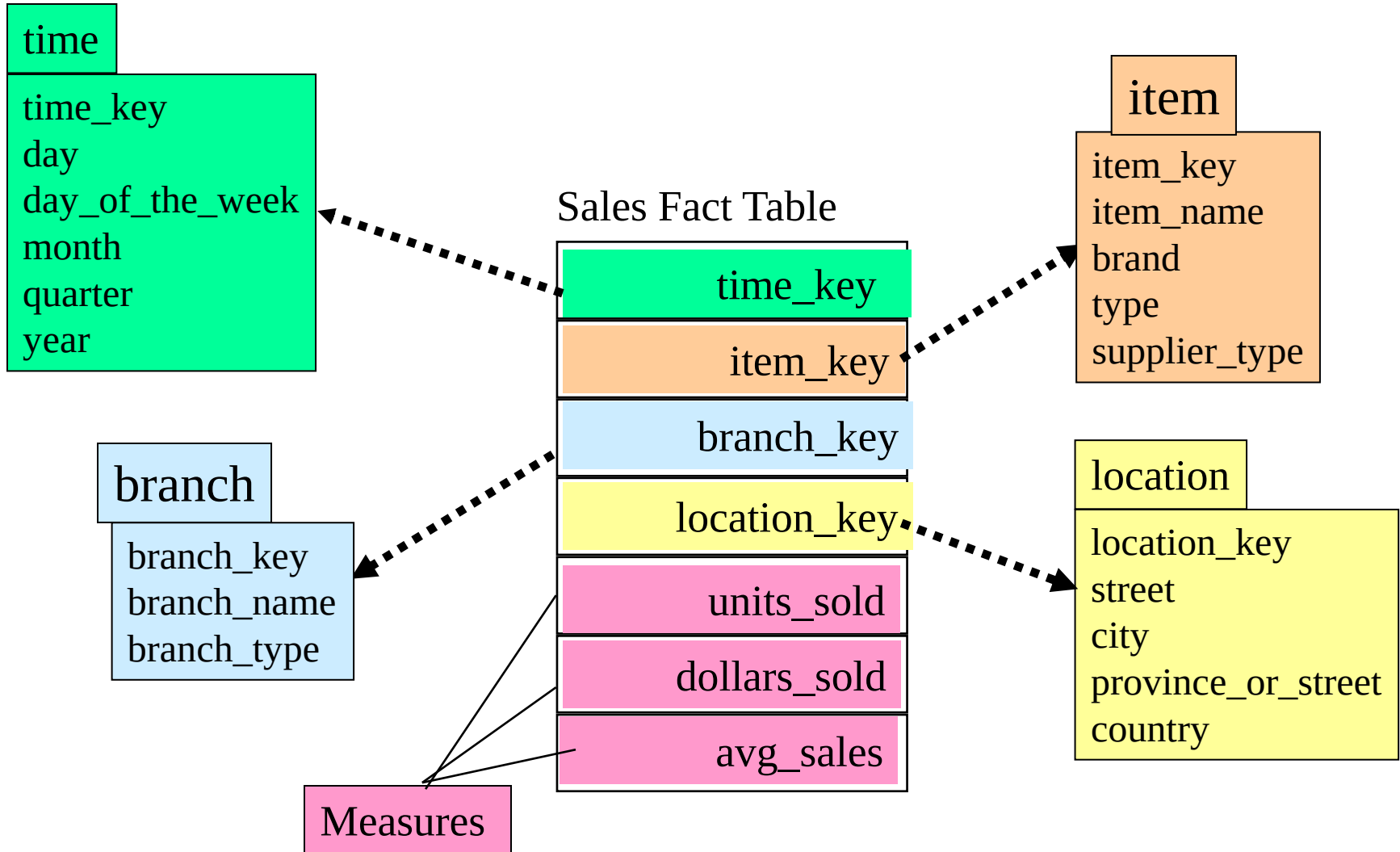
Cube: A Lattice of Cuboids



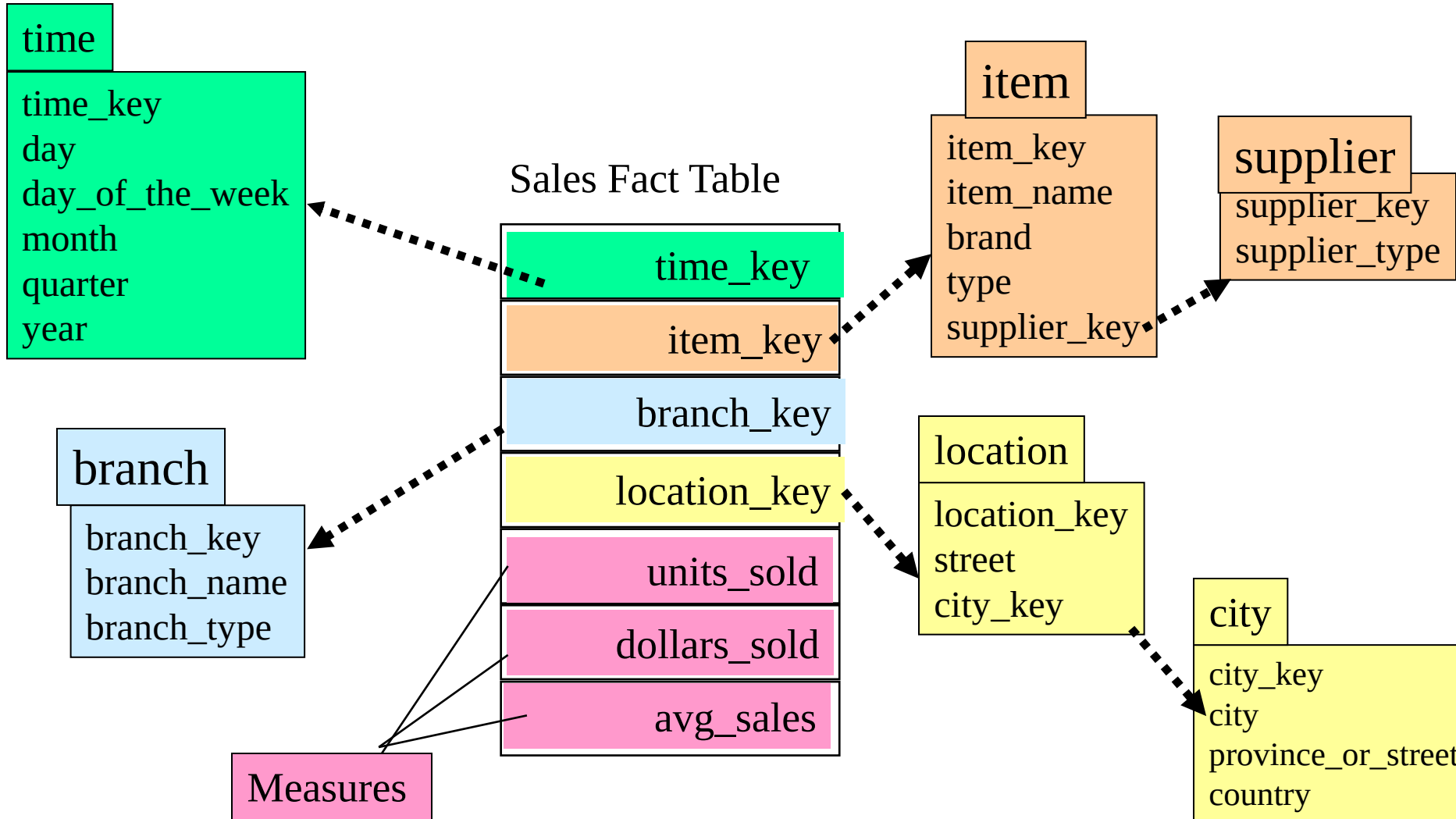
Conceptual Modeling of Data Warehouses

- Modeling data warehouses: dimensions & measures
 - Star schema: A fact table in the middle connected to a set of dimension tables
 - Snowflake schema: A refinement of star schema where some dimensional hierarchy is **normalized** into a set of smaller dimension tables, forming a shape similar to snowflake
 - Fact constellations: Multiple fact tables share dimension tables, viewed as a collection of stars, therefore called **galaxy schema** or fact constellation

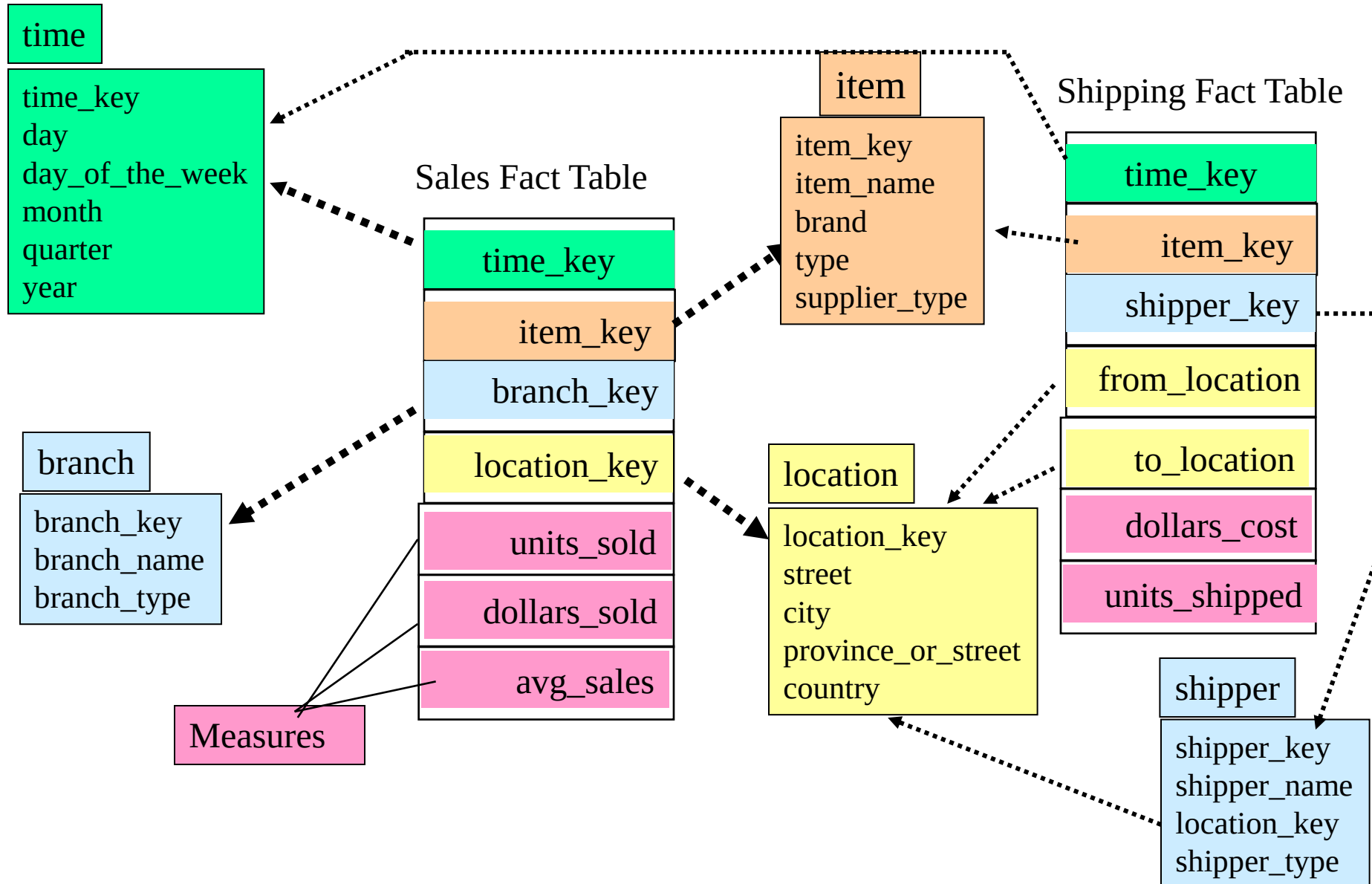
Example of Star Schema



Example of Snowflake Schema



Example of Fact Constellation



A Data Mining Query Language, DMQL: Language Primitives

- Cube Definition (Fact Table)

define cube <cube_name> [<dimension_list>]:
 <measure_list>

- Dimension Definition (Dimension Table)

define dimension <dimension_name> **as**
 (<attribute_or_subdimension_list>)

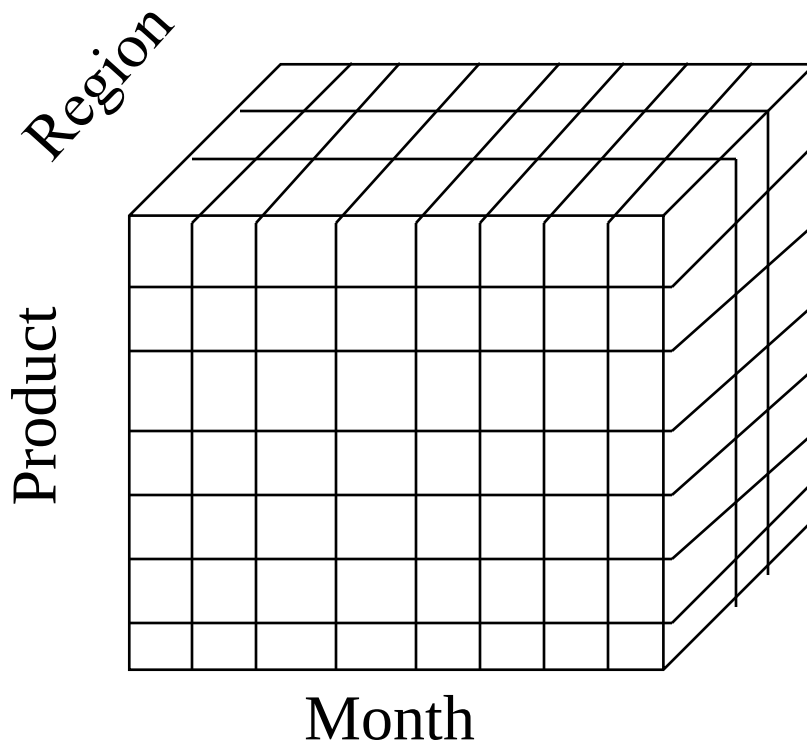
- Special Case (Shared Dimension Tables)

- First time as "cube definition"

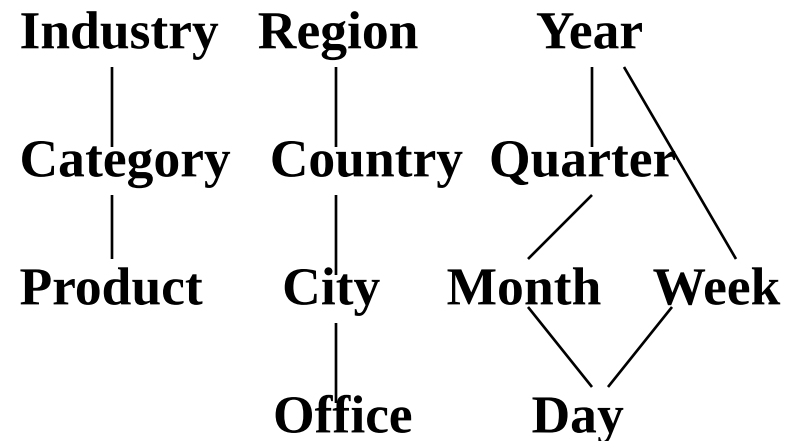
- **define dimension** <dimension_name> **as**
 <dimension_name_first_time> **in cube**
 <cube_name_first_time>

Multidimensional Data

- Sales volume as a function of product, month, and region



Dimensions: Product, Location, Time
Hierarchical summarization paths



A Sample Data Cube

